Exercise 5.8

Is C_2 uniquely decodeable?

Solution:

Yes, because we can determine all the position where 101 is from where 0s are. Then, we can determine that the remaining 1s are coded from 1s. The codewords can be uniquely decomposed and decoded.

Exercise 5.14

Solution:

Exercise 5.16

Solution:

Exercise 5.19

Is the code $\{00, 11, 0101, 111, 1010, 100100, 0110\}$ uniquely decodeable?

Solution:

No, since '111111' can be decoded as '11/11/11' or '111/111'.

Exercise 5.20

Is the ternary code $\{00, 012, 0110, 0112, 100, 201, 212, 22\}$ uniquely decodeable?

Solution:

This is a prefix code so yes, it is uniquely decodable.

Exercise

Solution: